

Analyse 1 – Exam Summary (Chapters 1–5)

L1 Informatique

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1 Chapter 1 – Field of Real Numbers

1.1 Core definitions and notations

Definition

Usual sets. $\mathbb{N} = \{0, 1, 2, \dots\}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$ (reals). Decimals: $\mathbb{D} = \left\{ \frac{p}{10^k} \in \mathbb{Q} : p \in \mathbb{Z}, k \in \mathbb{N} \right\}$.

Definition

Intervals. For $a \leq b$:

$$(a, b) = \{x \in \mathbb{R} : a < x < b\}, \quad [a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$$

and similarly $[a, b)$, $(a, b]$, (a, ∞) , $[a, \infty)$, $(-\infty, b)$, $(-\infty, b]$.

Definition

Absolute value.

$$|x| = \begin{cases} x & (x > 0) \\ 0 & (x = 0) \\ -x & (x < 0) \end{cases} \quad \text{and} \quad |x - a| < \varepsilon \iff a - \varepsilon < x < a + \varepsilon.$$

Definition

Integer part (floor). $E(x) \in \mathbb{Z}$ such that $E(x) \leq x < E(x) + 1$.

Definition

Bounds. Let $A \subset \mathbb{R}$.

- M is an **upper bound** if $\forall x \in A, x \leq M$.
- m is a **lower bound** if $\forall x \in A, x \geq m$.
- $\sup(A)$ is the **least** upper bound; $\inf(A)$ is the **greatest** lower bound.
- $\max(A)$ exists if $\sup(A) \in A$; $\min(A)$ exists if $\inf(A) \in A$.

1.2 Axioms and key theorems

Theorem / Proposition

Field + order. $\mathbb{R}$ is a commutative field and a totally ordered set, compatible with $+$ and $\times$.

Theorem / Proposition

Completeness. Every non-empty subset $A \subset \mathbb{R}$ bounded above admits $\sup(A)$ (and similarly $\inf(A)$ if bounded below).

Theorem / Proposition

Supremum characterization. $M = \sup(A)$ iff:

- (Upper bound) $\forall x \in A, x \leq M$,
- (ε -approximation) $\forall \varepsilon > 0, \exists x^* \in A : M - \varepsilon < x^*$.

(Analogous characterization for $\inf(A)$.)

Theorem / Proposition

Archimedean property. $\forall x \in \mathbb{R}, \exists n \in \mathbb{N} : n > x$.

Theorem / Proposition

Density of $\mathbb{Q}$ in $\mathbb{R}$. If $x < y$ in $\mathbb{R}$, then $\exists q \in \mathbb{Q}$ such that $x < q < y$.

1.3 Standard inequalities (used everywhere)

Theorem / Proposition

Triangle inequalities.

$$|x + y| \leq |x| + |y|, \quad ||x| - |y|| \leq |x + y|.$$

1.4 Proof skeletons (exam-ready)

Method Template

Proving $\sup(A) = M$ (template).

- Step 1: Prove $\forall x \in A, x \leq M$ (so M is an upper bound).
- Step 2: Let $\varepsilon > 0$. Find $x^* \in A$ such that $M - \varepsilon < x^*$.
- Step 3: Conclude by the supremum characterization.

Method Template

Density proof (template). For $x < y$:

- Choose $n \in \mathbb{N}$ such that $n(y - x) > 1$ (Archimedean).
- Let $p = E(nx) + 1$ so $nx < p \leq nx + 1 < ny$.
- Then $q = \frac{p}{n} \in \mathbb{Q}$ and $x < q < y$.

Common Traps

- $\sup(A)$ may exist without $\max(A)$ (e.g. $(0, 1)$ has $\sup = 1$ but no maximum).
- For negatives, $E(-1.5) = -2$ (not -1).
- In ε -proofs: keep the quantifiers order: $\forall \varepsilon > 0 \exists (\dots)$.

2 Chapter 2 – Field of Complex Numbers

2.1 Core definitions

Definition

Complex number. $z = a + ib$ with $a, b \in \mathbb{R}$ and $i^2 = -1$. Real and imaginary parts: $\Re(z) = a$, $\Im(z) = b$.

Definition

Conjugate and modulus. $\bar{z} = a - ib$, $|z| = \sqrt{a^2 + b^2}$.

Definition

Argument and trig/exponential form. For $z \neq 0$, choose θ such that

$$z = |z|(\cos \theta + i \sin \theta) = |z|e^{i\theta}.$$

(Argument is defined modulo 2π .)

2.2 Algebraic properties

Theorem / Proposition

Operations. For $z_1 = a + ib$ and $z_2 = c + id$:

$$z_1 + z_2 = (a + c) + i(b + d), \quad z_1 z_2 = (ac - bd) + i(ad + bc).$$

If $z_2 \neq 0$, division is obtained by multiplying numerator and denominator by $\bar{z}_2$.

Theorem / Proposition

Conjugate rules.

$$\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2, \quad \overline{z_1 z_2} = \bar{z}_1 \bar{z}_2, \quad z \bar{z} = |z|^2.$$

Theorem / Proposition

Modulus rules.

$$|z| \geq 0, \quad |z| = 0 \iff z = 0, \quad |z_1 z_2| = |z_1| |z_2|, \quad |z_1 + z_2| \leq |z_1| + |z_2|.$$

2.3 Moivre and roots

Theorem / Proposition

Euler (if used). $e^{it} = \cos t + i \sin t$.

Theorem / Proposition

Moivre. For $n \in \mathbb{N}$:

$$(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta).$$

Theorem / Proposition

n -th roots. If $z = |z|e^{i\theta}$ and $n \geq 1$, then the n distinct roots are

$$w_k = \sqrt[n]{|z|} e^{i\frac{\theta+2k\pi}{n}}, \quad k = 0, 1, \dots, n-1.$$

Method Template

Compute z^n (template).

- Convert z to $re^{i\theta}$.
- Compute $z^n = r^n e^{in\theta}$.
- Convert back to $a + ib$ if required.

Common Traps

- No order on $\mathbb{C}$ (“ $>$ ” and “ $<$ ” are undefined).
- $\Re(z)$ is the real number b , not ib .
- For roots: you must add $2k\pi$; forgetting it loses solutions.

3 Chapter 3 – Sequences of Real Numbers

3.1 Definitions

Definition

A real sequence is a function $u : \mathbb{N} \rightarrow \mathbb{R}$, written $(u_n)_{n \in \mathbb{N}}$.

Definition

Boundedness. Bounded above: $\exists M \in \mathbb{R}, \forall n, u_n \leq M$. Bounded below: $\exists m \in \mathbb{R}, \forall n, u_n \geq m$. Bounded: both.

Definition

Monotonicity. Increasing: $\forall n, u_{n+1} \geq u_n$. Decreasing: $\forall n, u_{n+1} \leq u_n$.

Definition

Limit. (u_n) converges to $l \in \mathbb{R}$ if

$$\forall \varepsilon > 0, \exists n_0 \in \mathbb{N}, \forall n \geq n_0 : |u_n - l| < \varepsilon.$$

Definition

Divergence to $\pm\infty$.

$$u_n \rightarrow +\infty \iff \forall A > 0, \exists n_0, \forall n \geq n_0 : u_n > A,$$

and similarly for $-\infty$.

Definition

Cauchy. (u_n) is Cauchy if $\forall \varepsilon > 0, \exists n_0, \forall p, q \geq n_0 : |u_p - u_q| < \varepsilon$.

Definition

Subsequence. $(u_{n_k})_{k \in \mathbb{N}}$ where (n_k) is strictly increasing. A **cluster point** is a limit of some subsequence.

Definition

Adjacent sequences. (u_n) and (v_n) are *adjacent* if:

- (u_n) is increasing, (v_n) is decreasing,
- $\forall n, u_n \leq v_n$,
- $v_n - u_n \rightarrow 0$.

3.2 Main theorems

Theorem / Proposition

Uniqueness of the limit. If $\lim u_n = l_1$ and $\lim u_n = l_2$, then $l_1 = l_2$.

Theorem / Proposition

Convergent $\Rightarrow$ bounded. If (u_n) converges, then it is bounded.

Theorem / Proposition

Algebra of limits. If $u_n \rightarrow l$ and $v_n \rightarrow m$, then:

$$u_n + v_n \rightarrow l + m, \quad u_n v_n \rightarrow lm, \quad \frac{u_n}{v_n} \rightarrow \frac{l}{m} \text{ if } v_n \neq 0 \text{ eventually and } m \neq 0.$$

Theorem / Proposition

Order and squeeze. If $u_n \leq v_n \leq w_n$ eventually and $u_n \rightarrow l$, $w_n \rightarrow l$, then $v_n \rightarrow l$.

Theorem / Proposition

Monotone convergence. Increasing and bounded above $\Rightarrow$ convergent. Decreasing and bounded below $\Rightarrow$ convergent.

Theorem / Proposition

Adjacent sequences. If (u_n) and (v_n) are adjacent, then both converge and have the same limit.

Theorem / Proposition

Cauchy criterion (in $\mathbb{R}$). A sequence is convergent iff it is Cauchy.

Theorem / Proposition

Bolzano–Weierstrass. Every bounded sequence has a convergent subsequence.

3.3 Proof skeletons

Method Template

Uniqueness (template). Assume $l_1 \neq l_2$, choose $\varepsilon = \frac{|l_1 - l_2|}{4}$, use triangle inequality to reach a contradiction.

Method Template

Monotone + bounded (template). Let $M = \sup\{u_n\}$ (or $m = \inf\{u_n\}$). Use the ε -characterization of sup to show (u_n) is Cauchy, hence convergent.

Common Traps

- Do not forget “eventually” in inequalities: there exists n_0 such that for all $n \geq n_0$.
- Monotone does not imply bounded: you need both for monotone convergence.
- Existence of subsequences: “two subsequences to two different limits” implies divergence of the original sequence.

4 Chapter 4 – Limits and Continuous Functions

4.1 Core definitions

Definition

Numerical function. A function $f : E \subset \mathbb{R} \rightarrow \mathbb{R}$. Domain: $D_f = E$. Graph: $\Gamma_f = \{(x, f(x)) : x \in D_f\} \subset \mathbb{R}^2$.

Definition

Monotonicity, parity, periodicity.

- Increasing: $\forall x \leq y$ in D_f , $f(x) \leq f(y)$ (strict: $<$).
- Even: $x \in D_f \Rightarrow -x \in D_f$ and $f(-x) = f(x)$. Odd: $f(-x) = -f(x)$.
- Periodic (period $p > 0$): $x \in D_f \Rightarrow x + p \in D_f$ and $f(x + p) = f(x)$.

Definition

Limit at a point (epsilon–delta). $\lim_{x \rightarrow x_0} f(x) = \ell$ means:

$$\forall \varepsilon > 0, \exists \delta > 0, \forall x \in D_f : 0 < |x - x_0| < \delta \Rightarrow |f(x) - \ell| < \varepsilon.$$

Definition

One-sided limits. $\lim_{x \rightarrow x_0^-} f(x)$ and $\lim_{x \rightarrow x_0^+} f(x)$ are defined by restricting $x < x_0$ or $x > x_0$.

Definition

Infinite limits / limits at infinity.

$$\lim_{x \rightarrow x_0} f(x) = +\infty \iff \forall A > 0, \exists \delta > 0 : 0 < |x - x_0| < \delta \Rightarrow f(x) > A,$$

$$\lim_{x \rightarrow +\infty} f(x) = \ell \iff \forall \varepsilon > 0, \exists B > 0 : x > B \Rightarrow |f(x) - \ell| < \varepsilon$$

(and similarly for $-\infty$).

Definition

Continuity. f is continuous at $x_0 \in D_f$ iff $\lim_{x \rightarrow x_0} f(x) = f(x_0)$. A **continuous extension** at $x_0 \notin D_f$ is defined by $f(x_0) := \lim_{x \rightarrow x_0} f(x)$ (if the limit exists).

4.2 Main theorems

Theorem / Proposition

Uniqueness of limit. If $\lim_{x \rightarrow x_0} f(x)$ exists, it is unique.

Theorem / Proposition

Two-sided limit vs one-sided limits.

$$\lim_{x \rightarrow x_0} f(x) = \ell \iff \lim_{x \rightarrow x_0^-} f(x) = \ell \text{ and } \lim_{x \rightarrow x_0^+} f(x) = \ell.$$

Theorem / Proposition**Sequential characterization (limit).**

$$\lim_{x \rightarrow x_0} f(x) = \ell \iff \forall (x_n) \subset D_f, x_n \rightarrow x_0, x_n \neq x_0, f(x_n) \rightarrow \ell.$$

Theorem / Proposition**Algebra of limits.** If $\lim f = \ell$ and $\lim g = m$ at x_0 , then:

$$\lim(f \pm g) = \ell \pm m, \quad \lim(fg) = \ell m, \quad \lim \frac{f}{g} = \frac{\ell}{m} \text{ if } m \neq 0 \text{ and } g(x) \neq 0 \text{ near } x_0.$$

Theorem / Proposition**Composition.** If $f(x) \rightarrow y_0$ as $x \rightarrow x_0$ and $g(u) \rightarrow \ell$ as $u \rightarrow y_0$, then $g(f(x)) \rightarrow \ell$ as $x \rightarrow x_0$.**Theorem / Proposition****Continuity – operations.** Sums, products, scalar multiples, quotients (denominator nonzero) of continuous functions are continuous.**Theorem / Proposition****Sequential characterization (continuity).** f is continuous at x_0 iff for every $(x_n) \subset D_f$ with $x_n \rightarrow x_0$, we have $f(x_n) \rightarrow f(x_0)$.**Theorem / Proposition****Non-vanishing near a point.** If f is continuous at x_0 and $f(x_0) \neq 0$, then $\exists$ a neighborhood of x_0 where $f(x) \neq 0$.**Theorem / Proposition****Weierstrass (Extreme Value).** If f is continuous on $[a, b]$, then f is bounded and attains its minimum and maximum on $[a, b]$.**Theorem / Proposition****Bolzano (Root existence).** If f is continuous on $[a, b]$ and $f(a)f(b) \leq 0$, then $\exists c \in [a, b]$ such that $f(c) = 0$.**Theorem / Proposition****Intermediate Value Property.** If f is continuous on $[a, b]$, then for every y between $f(a)$ and $f(b)$, there exists $c \in [a, b]$ such that $f(c) = y$. In particular, the image of an interval by a continuous function is an interval.**Theorem / Proposition****Strict monotonicity and inverse.** If f is continuous and strictly monotone on an interval I , then:

- f is injective and induces a bijection $f : I \rightarrow f(I)$,
- the inverse $f^{-1} : f(I) \rightarrow I$ is continuous.

4.3 Method library (exam mode)

Method Template

Epsilon–delta limit (linear/polynomial).

- Goal: make $|f(x) - \ell| < \varepsilon$ from a bound on $|x - x_0|$.
- Common move: factor $|x - x_0|$ and bound the remaining factor using “ $|x - x_0| < 1$ ”.
- Choose $\delta = \min(1, \varepsilon/C)$ where C is the bound you obtained.

Method Template

Continuity / extension at x_0 .

- Compute $\lim_{x \rightarrow x_0} f(x)$ (simplify/factor/rationalize).
- If the limit equals $f(x_0)$: continuous.
- If $x_0 \notin D_f$ and the limit exists: define the extension $\tilde{f}(x_0)$ equal to that limit.

Method Template

Bolzano (find a root).

- Check continuity on $[a, b]$.
- Compute $f(a)$ and $f(b)$.
- If $f(a)f(b) \leq 0$, conclude $\exists c$ with $f(c) = 0$.

Common Traps

- A two-sided limit exists *only if* left and right limits exist and are equal.
- Always state the domain: limits/continuity require x in D_f and a punctured neighborhood when needed.
- Indeterminate forms ($\infty - \infty$, $0 \cdot \infty$, ∞/∞ , $0/0$) are not automatic limits: you must transform the expression.

5 Chapter 5 – Differential Calculus (One Variable)

5.1 Core definitions

Definition

Derivative at a point. f is differentiable at $x_0 \in D_f$ if the limit exists:

$$f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}.$$

Definition

One-sided derivatives.

$$f'_+(x_0) = \lim_{h \rightarrow 0^+} \frac{f(x_0 + h) - f(x_0)}{h}, \quad f'_-(x_0) = \lim_{h \rightarrow 0^-} \frac{f(x_0 + h) - f(x_0)}{h}.$$

Definition

Higher derivatives and classes. $f'' = (f')'$, and more generally $f^{(n)}$. $C^n(I)$: n times differentiable with $f^{(n)}$ continuous. $C^\infty(I)$: $C^n(I)$ for all n .

5.2 Derivative properties

Theorem / Proposition

Differentiability implies continuity. If f is differentiable at x_0 , then f is continuous at x_0 .

Theorem / Proposition

Rules. If f and g are differentiable at x_0 :

$$(f \pm g)'(x_0) = f'(x_0) \pm g'(x_0), \quad (fg)'(x_0) = f'(x_0)g(x_0) + f(x_0)g'(x_0).$$

If $g(x_0) \neq 0$ (and nonzero near x_0): $\left(\frac{f}{g}\right)'(x_0) = \frac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{g(x_0)^2}.$

Theorem / Proposition

Chain rule. If f differentiable at x_0 and g differentiable at $f(x_0)$:

$$(g \circ f)'(x_0) = g'(f(x_0)) f'(x_0).$$

Theorem / Proposition

Derivative of the inverse (under hypotheses). If f is bijective, differentiable at x_0 , and $f'(x_0) \neq 0$, then f^{-1} is differentiable at $y_0 = f(x_0)$ and:

$$(f^{-1})'(y_0) = \frac{1}{f'(x_0)}.$$

5.3 Rolle and Mean Value Theorems

Theorem / Proposition

Rolle. If f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then $\exists c \in (a, b)$ such that $f'(c) = 0$.

Theorem / Proposition

Mean Value Theorem (MVT). If f is continuous on $[a, b]$ and differentiable on (a, b) , then $\exists c \in (a, b)$ such that:

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Theorem / Proposition

Generalized MVT (Cauchy). If f, g are continuous on $[a, b]$ and differentiable on (a, b) , then $\exists c \in (a, b)$:

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)} \quad (\text{when } g(b) \neq g(a)).$$

5.4 Applications

Theorem / Proposition

Monotonicity via derivative (typical).

- If $f'(x) > 0$ on an interval I , then f is strictly increasing on I .
- If $f'(x) = 0$ on I , then f is constant on I .

Theorem / Proposition

L'Hôpital (0/0 and ∞/∞). If the hypotheses stated in the course hold (differentiability on a punctured neighborhood, $g'(x) \neq 0$ near the point, and a suitable form 0/0 or ∞/∞), then:

$$\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)}$$

provided the right-hand limit exists (finite or infinite).

Theorem / Proposition

Leibniz (product n -th derivative).

$$(fg)^{(n)} = \sum_{k=0}^n \binom{n}{k} f^{(n-k)} g^{(k)}.$$

Theorem / Proposition

Taylor–Lagrange (order n). If f is $(n+1)$ times differentiable on an interval containing x and x_0 , then:

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + \frac{f^{(n+1)}(c)}{(n+1)!} (x - x_0)^{n+1}$$

for some c between x and x_0 . (Maclaurin is the case $x_0 = 0$.)

5.5 Proof skeletons

Method Template

MVT proof (template). Define $g(x) = f(x) - f(a) - \frac{f(b)-f(a)}{b-a}(x-a)$ so $g(a) = g(b) = 0$. Apply Rolle to get $g'(c) = 0$ and conclude.

Method Template

Use MVT to prove an inequality (template).

- Identify an interval $[a, b]$ and a function f whose increment encodes the inequality.
- Apply MVT: $f(b) - f(a) = f'(c)(b - a)$.
- Bound $f'(c)$ using known estimates, then conclude.

Common Traps

- Rolle needs $f(a) = f(b)$; MVT does not.
- Chain rule: $(g \circ f)' = g'(f(x))f'(x)$ (not $g'(x)f'(x)$).
- L'Hôpital: never apply before checking hypotheses (form + differentiability + $g' \neq 0$).
- Differentiable implies continuous, but continuous does *not* imply differentiable (e.g. $|x|$ at 0).

Exam Checklist

- **Before using MVT/Rolle:** continuity on $[a, b]$, differentiability on (a, b) .
- **Before using inverse derivative:** bijective + differentiable at x_0 + $f'(x_0) \neq 0$.
- **Before using L'Hôpital:** correct form ($0/0$ or ∞/∞) + hypotheses + compute $\frac{f'}{g'}$.
- **Taylor:** identify x_0 , order n , write polynomial + remainder, then bound remainder if needed.